25 Expected Values of Linear Combinations of Random Variables

A **linear rescaling** of a random variable X is of the form $aX+b$ where a and b are non-random constants. A **linear combination** of two random variables X and Y is of the form $aX + bY$ where a and b are non-random constants.

Properties of expected value

- Expected value and linear rescaling: If X is a random variable and a, b are non-random constants then
$$\mathrm{E}(aX + b) = a\mathrm{E}(X) + b$$
- Linearity of expected value: For *any* two random variables X and Y,
$$\mathrm{E}(X + Y) = \mathrm{E}(X) + \mathrm{E}(Y)$$
- That is, the expected value of the sum is the sum of expected values, regardless of how the random variables are related.
- Therefore, you only need to know the marginal distributions of X and Y to find the expected value of their sum. (But keep in mind that the *distribution* of $X + Y$ will depend on the joint distribution of X and Y.)
- Combining properties of linear rescaling with linearity of expected value yields the expected value of a linear combination, which extends naturally to more than two random variables.
$$\mathrm{E}(aX + bY + c) = a\mathrm{E}(X) + b\mathrm{E}(Y) + c$$

Properties of variance

- Variance and linear rescaling: If X is a random variable and a, b are non-random constants then
$$\mathrm{SD}(aX + b) = |a|\mathrm{SD}(X)$$
$$\mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$$
- Variance of sums and differences of random variables
$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X, Y)$$
$$\mathrm{Var}(X - Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) - 2\mathrm{Cov}(X, Y)$$

- The variance of the sum (or difference) is the sum of the variances if and only if X and Y are uncorrelated.

$$\begin{aligned}
\mathrm{Var}(X+Y) &= \mathrm{Var}(X) + \mathrm{Var}(Y) \qquad \text{if } X, Y \text{ are uncorrelated}\\
\mathrm{Var}(X-Y) &= \mathrm{Var}(X) + \mathrm{Var}(Y) \qquad \text{if } X, Y \text{ are uncorrelated}
\end{aligned}$$

- If a, b, c are non-random constants and X and Y are random variables then

$$\mathrm{Var}(aX + bY + c) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y) + 2ab\mathrm{Cov}(X, Y)$$

Example 25.1. Recall the SAT score Application assignment.

Let M be the Math score and R be the Reading (and Writing) score of a randomly selected SAT taker. We'll assume (M, R) pairs follow a *Bivariate Normal distribution*.

- Math scores (M) have mean 500 and standard deviation 140
- Reading scores (R) have mean 520 and standard deviation 110

For each of the following scenarios, compute

- $\mathrm{E}(M + R)$
- $\mathrm{Var}(M + R)$
- $\mathrm{SD}(M + R)$

1. $\mathrm{Corr}(M, R) = 0$

2. $\mathrm{Corr}(M, R) = 0.6$

3. $\mathrm{Corr}(M, R) = -0.9$

4. In which of these scenarios is $\text{SD}(M + R)$ largest? Smallest? Can you explain why intuitively?

Example 25.2. Recall the SAT score Application assignment.

Let M be the Math score and R be the Reading (and Writing) score of a randomly selected SAT taker. We'll assume (M, R) pairs follow a *Bivariate Normal distribution*.

- Math scores (M) have mean 500 and standard deviation 140
- Reading scores (R) have mean 520 and standard deviation 110

For each of the following scenarios, compute

- $\text{E}(M - R)$
- $\text{Var}(M - R)$
- $\text{SD}(M - R)$

1. $\text{Corr}(M, R) = 0$

2. $\text{Corr}(M, R) = 0.6$

3. $\text{Corr}(M, R) = -0.9$

4. In which of these scenarios is $\text{SD}(M - R)$ largest? Smallest? Can you explain why intuitively?

- Two continuous random variables X and Y have a **Bivariate Normal** distribution with parameters μ_X, μ_Y, $\sigma_X > 0$, $\sigma_Y > 0$, and $-1 < \rho < 1$ if the joint pdf is

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right)\right]\right.$$

- It can be shown that if the pair (X, Y) has a BivariateNormal(μ_X, μ_Y, σ_X, σ_Y, ρ) distribution

$$\begin{aligned}
\mathrm{E}(X) &= \mu_X \\
\mathrm{E}(Y) &= \mu_Y \\
\mathrm{SD}(X) &= \sigma_X \\
\mathrm{SD}(Y) &= \sigma_Y \\
\mathrm{Corr}(X,Y) &= \rho
\end{aligned}$$

- A Bivariate Normal Density has elliptical contours. For each height $c > 0$ the set $\{(x, y) : f_{X,Y}(x, y) = c\}$ is an ellipse. The density decreases as (x, y) moves away from (μ_X, μ_Y), most steeply along the minor axis of the ellipse, and least steeply along the major of the ellipse.
- A scatterplot of (x, y) pairs generated from a Bivariate Normal distribution will have a rough linear association and the cloud of points will resemble an ellipse.
- If X and Y have a Bivariate Normal distribution, then the marginal distributions are also Normal: X has a Normal(μ_X, σ_X) distribution and Y has a Normal(μ_Y, σ_Y).
- If X and Y have a Bivariate Normal distribution and $\mathrm{Corr}(X, Y) = 0$ then X and Y are independent. (Remember, in general it is possible to have situations where the correlation is 0 but the random variables are not independent.)
- **X and Y have a Bivariate Normal distribution if and only if every linear combination of X and Y has a Normal distribution.** That is, X and Y have a Bivariate Normal distribution if and only if $aX + bY$ has a Normal distribution for all a, b.

Example 25.3. Let M be the Math score and R be the Reading (and Writing) score of a randomly selected SAT taker. We'll assume (M, R) pairs follow a *Bivariate Normal distribution.*

- Math scores (M) have mean 500 and standard deviation 140
- Reading scores (R) have mean 520 and standard deviation 110
- The correlation is 0.6

1. Compute the probability that a student has a total score above 1400.

2. Compute the probability that a student has a higher Math than Reading score.

Properties of covariance

$$\begin{aligned}
\mathrm{Cov}(X,X) &= \mathrm{Var}(X) \\
\mathrm{Cov}(X,Y) &= \mathrm{Cov}(Y,X) \\
\mathrm{Cov}(X,c) &= 0 \\
\mathrm{Cov}(aX+b, cY+d) &= ac\mathrm{Cov}(X,Y) \\
\mathrm{Cov}(X+Y,\ U+V) &= \mathrm{Cov}(X,U)+\mathrm{Cov}(X,V)+\mathrm{Cov}(Y,U)+\mathrm{Cov}(Y,V)
\end{aligned}$$

Example 25.4. Let X be the number of two-point field goals a basketball player makes in a game, Y the number of three point field goals made, and Z the number of free throws made (worth one point each). Assume X, Y, Z have standard deviations of 2.5, 3.7, 1.8, respectively, and $\mathrm{Corr}(X,Y) = 0.1$, $\mathrm{Corr}(X,Z) = 0.3$, $\mathrm{Corr}(Y,Z) = -0.5$.

1. Find the standard deviation of the number of fields goals in a game (not including free throws)

2. Find the standard deviation of total points scored on fields goals in a game (not including free throws)

3. Find the standard deviation of total points scored in a game.